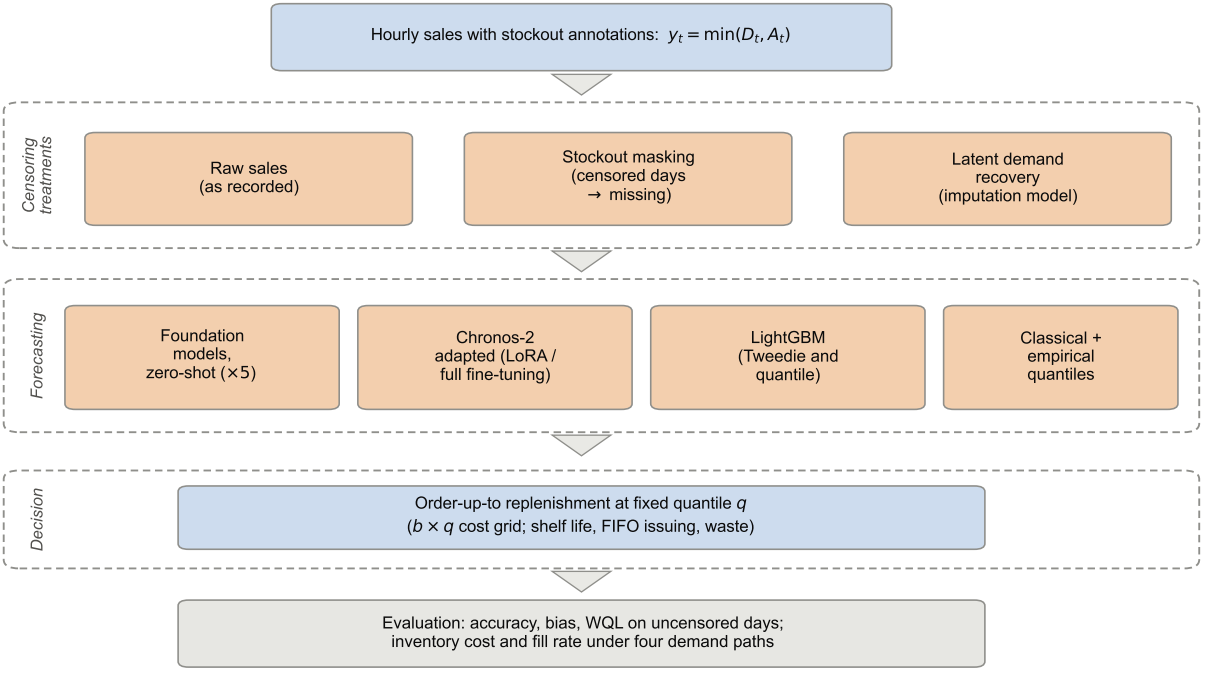
Hourly sales with stockout annotations: $y_t = \min(D_t, A_t)$
Censoring treatments
Raw sales (as recorded)
Stockout masking (censored days → missing)
Latent demand recovery (imputation model)
Forecasting
Foundation models, zero-shot (×5)
Chronos-2 adapted (LoRA / full fine-tuning)
LightGBM (Tweedie and quantile)
Classical + empirical quantiles
Decision
Order-up-to replenishment at fixed quantile q ($b \times q$ cost grid; shelf life, FIFO issuing, waste)
Evaluation: accuracy, bias, WQL on uncensored days; inventory cost and fill rate under four demand paths